

ECHOES OF MARSILI

An Interactive Hybrid Narrative Game for Heritage, Science, and Discovery

PROJECT DESIGN BRIEF

Revised Edition — Grounded in CoDesignTool Outcomes

Palazzo Poggi Museum · Marsili Collection · Bologna, Italy
Museum and Heritage Interactive Media Design · 2025–2026

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0. How This Brief Was Produced

This design brief is not written from intuition. It is the direct outcome of a structured co-design session carried out with the CoDesignTool (the printable VisitorBox card deck developed under the EU PERCEIVE project). Over seven staged rounds, the team played and ranked a set of decision cards, and the choices that survived each round now form the backbone of every section below.

In short: every requirement, goal, and feature in this document can be traced back to a card the team consciously selected. The brief therefore records both what we decided and why, so the rationale stays visible to the whole team and to our reviewers.

The seven design stages we played through:

- 1. Context** — who we are designing for and which assets matter most.
- 2. Institutional Goals** — what the museum itself wants to gain.
- 3. Audience Goals** — the emotional and cognitive change we want in visitors (Authenticity sub-deck).
- 4. Design Brief** — concrete requirements: motivations, barriers, capabilities, devices.
- 5. Ideation** — translating requirements into features (technology, engagement, audience-goal strategies).
- 6. Disruption** — stress-testing the idea against real threats.
- 7. Storyboard** — sequencing the experience from first contact to conclusion.

The remainder of this brief presents the resolved design. Where a section reflects a specific card decision, this is noted so the link between play and product remains traceable.

1. Project Overview

The defining ambition of the project, fixed during the Context stage of our co-design session, is reach: we are designing for someone who arrives knowing nothing about Marsili and leaves knowing as much about him as a single visit can convey. The experience must therefore carry a complete newcomer from zero to genuine understanding, without ever assuming prior knowledge.

The project shifts visitors from passive observers into active participants. Rather than reading labels and moving on, visitors make choices, adopt a role, listen to narrated scenes, solve light tasks, and assemble Marsili's story for themselves as they move through the rooms.

2. Context (Stage 1)

2.1 Target Audience

Our Context cards defined an audience the institution does not yet reliably attract and would like to win over. We expressed this as a clear design target rather than a demographic label.

Audience card (as played):

"Someone with no knowledge of Marsili who, after our experience, knows as much about him as they can."

Combining this card with our project notes, the primary audience is young people aged 14 to 27 — secondary-school and university students, and culturally curious young adults — who are not current visitors of the Marsili Collection. Crucially, no prior familiarity with Marsili or with seventeenth- and eighteenth-century science is assumed. The experience is built to bring a newcomer fully up to speed.

This audience is comfortable with everyday digital media: they play games, browse websites, and in some cases build web content themselves. This shaped the capability assumptions recorded later in the brief.

2.2 Star Assets

During the Context stage we also selected the venue's strongest selling points — the assets around which the experience should be anchored.

Star Asset	Why it matters to the experience
Maps	Marsili's cartographic work, including his surveys of the Danube and the Bosphorus, anchors the geography and discovery chapters of the story.

Star Asset	Why it matters to the experience
Armenian dome / architecture	A distinctive architectural and cultural landmark that grounds the experience in a specific, recognisable place.
The historical city	Bologna itself — its scientific heritage and Marsili's role in founding the Institute of Sciences — frames the wider narrative.

These assets are not backdrops. They are the narrative anchors: each becomes a point in the museum where a chapter of Marsili's story is unlocked and explored.

3. Institutional Goals (Stage 2)

Beyond the visitor, the museum has its own objectives. From the Institutional Goals deck the team selected and ranked the three goals the experience should serve.

Institutional Goal	Rank	How the experience delivers it
Higher visitor satisfaction	Primary	Visitors should leave feeling the visit was worthwhile, engaging, and memorable — the headline measure of the project's success.
Increase educational activities	Secondary	The experience adds a genuinely educational, interactive layer to a collection that currently relies largely on static panels.
Digitise more assets	Supporting	Building the Twine narrative and its media drives the digitisation of maps, instruments, and archival materials into reusable digital form.

These goals are mutually reinforcing. A more satisfying, educational visit raises engagement; producing it requires digitising assets that the museum can then reuse well beyond this project.

4. Audience Goals (Stage 3)

At the Audience Goals stage we chose the Authenticity sub-deck: our priority is to make Marsili's world feel real, present, and personally resonant. The cards we selected fall into two of the deck's sub-categories — Self (the visitor's inner experience) and World (the surrounding atmosphere).

4.1 Authenticity — World

- **Let visitors' time flow** — the experience should absorb visitors so the visit feels immersive rather than rushed or clock-watched.
- **Create the appropriate mood** — visual style, sound, and lighting work together to evoke Marsili's eighteenth-century world.
- **Make visitors feel familiar with the environment** — visitors should quickly feel at home in the space and the story, never lost or excluded.

4.2 Authenticity — Self

- **Strengthen personal context** — the story connects to the visitor's own perspective, making Marsili's choices feel personally relevant.
- **Develop a sense of familiarity** — recurring narrative cues and a consistent guide (Marsili himself) build comfort and recognition over the visit.

- **Stimulate sensory responses** — sound, and where possible smell and touch, are engaged so the experience is felt, not just read.

4.3 Blending in Sense of Care

Although Authenticity was our primary lens, the ideas we generated later (Stage 5) drew on the Sense of Care approach — historical characters, enchantment, and theory-of-mind storytelling. We deliberately keep both: Authenticity governs how the world feels (mood, immersion, sensory presence), while Sense of Care governs how the visitor connects emotionally to the people in that world (empathy, wonder, understanding others' motivations). The two are complementary, and together they define an audience-centred emotional target — a believable world inhabited by people the visitor comes to care about.

5. Design Requirements (Stage 4)

The Design Brief stage converted our goals into concrete, testable requirements across four categories: motivations, barriers, capabilities, and devices.

5.1 Motivations — what draws visitors in

Motivation	Design implication
Time travel	Visitors are drawn by the chance to step into an unfamiliar but intriguing way of life — to experience Marsili's era from the inside rather than read about it from outside.
Entertainment	Visitors want to be amused, intrigued, and surprised. The experience should feel like a story and a game, not a lecture.

5.2 Barriers — what we must remove

Barrier	How the design responds
Poor signage	Many visitors cannot navigate or interpret a museum that relies on dense, single-language, academic labels. The experience must provide clear, large-print, plain-language, multilingual guidance and navigation.
Lack of access to technology	Visitors without a capable smartphone must not get a second-rate visit. The museum will provide preloaded backup tablets, and the core story must work on modest devices.

5.3 Capabilities — what we can assume visitors can do

- **Computer gaming** — the audience is comfortable with game conventions: choices, progression, simple puzzles.
- **Using websites** — visitors can confidently find and navigate a web-based experience.
- **Web content creation** — a portion of the audience can create and edit web content, supporting optional participatory or share-based features.

5.4 Devices — what the experience must run on

The experience is delivered through three device classes selected from the deck. It must run well on all of them, with no downloads or technical skill required.

Device	Role	Rationale
Smartphone	Primary	Internet- and GPS-connected, supports the story app, and lets visitors create and share content. The default personal device.

Device	Role	Rationale
Tablet	Shared / backup	Larger touchscreen that can be shared by small groups; the form provided by the museum as preloaded backup units.
Computer	Pre-visit / online	Laptop or desktop used to discover and play part of the experience online before arriving at the museum.

6. From Requirements to Features (Stage 5: Ideation)

With requirements fixed, the Ideation stage generated the actual features of the experience across three subcategories: technologies, engagement mechanics, and audience-goal strategies.

6.1 Technologies

Technology	How we use it
Audio	Marsili narrates his own story. We use AI-generated voice (ElevenLabs) plus ambient sound to carry the narrative and reduce reliance on reading.
Augmented authenticity	Sight, sound, and where feasible smell and touch are augmented — for example a coffee aroma cue or tactile artifact replicas — to make scenes feel genuinely present.
Online	Part of the experience lives online: it can be discovered and begun on a website before the visit, then continued in the museum.

These sit alongside the production toolchain recorded in our notes: Twine (SugarCube) for the branching narrative, Figma for interface design, HTML / CSS / JavaScript for the web layer, and GitHub Pages for offline-capable hosting.

6.2 Engagement Mechanics

Mechanic	What it does for the visitor
Decision	The visitor makes choices that shape what follows — for example whether Marsili leans on books or lived experience — giving a sense of authorship over the story.
Persona	The visitor adopts or constructs a character for the visit (framed through Caterina, who awakens Marsili's memory), viewing events through a personal lens rather than as a detached observer.

6.3 Audience-Goal Strategies (Sense of Care)

To turn engagement into genuine emotional connection, we drew on the Sense of Care ideation cards:

- **Historical characters** — a narrative that mixes documented fact with evocative storytelling builds a deep connection between visitor and heritage.
- **Enchantment / wonder** — each chapter is designed to spark a sense of wonder, whether at hidden ocean currents or debunked sea-monster myths.
- **Theory-of-mind stories** — visitors are invited to imagine the motivations, feelings, and intentions behind characters' actions, deepening empathy and reflection.

7. Stress-Testing the Design (Stage 6: Disruption)

Before committing to the storyboard, we played the Disruption deck to bulletproof the idea against realistic threats. Each card raised a scenario; the team agreed a mitigation.

Threat	Scenario	Our mitigation
Accessibility (Environment)	Visitors with different physical abilities may struggle to access parts of the experience.	Subtitles for all audio, adjustable text size, alt-text, screen-reader support, high-contrast visuals for low-light rooms, large tap targets, and step-free routing. Audio, text, visuals, and touch each carry the core content independently.
Attendance (Resilience)	Visitor numbers and connectivity may vary; the museum WiFi may be unreliable.	The Twine story is lightweight and runs offline once loaded. Preloaded museum tablets cover visitors without a capable device, keeping the experience robust under varying conditions.
Cognitive balance (Engagement)	The experience might tip too far toward being purely educational, purely fun, or too difficult.	Each chapter blends a learning objective, a playful element, and a light optional challenge. Tasks are skippable so no one gets stuck, keeping the experience balanced for a broad audience.

Where a threat exposed a weakness, we revisited earlier decisions and adjusted — exactly the iterative loop the CoDesignTool is built to encourage.

8. Experience Storyboard (Stage 7)

The final stage sequences the experience. Our project notes describe a flow that begins well before the visitor reaches the museum and ends with them inside the story. We express it in five moments.

8.1 The Five Moments

Moment	Where	What happens
1 • Discovery	Online / pre-visit	The experience is advertised through schools, the internet, and museum channels. A young person encounters it and can begin playing online — meeting Marsili and the premise before ever arriving.
2 • Arrival	Entrance Hall	The visitor (often a student, sometimes a class group) arrives at the Marsili Collection. Ambient sound and lighting establish mood; Marsili welcomes them and the on-site chapters unlock.
3 • Exploration	Through the rooms	Moving room to room, the visitor scans artifacts to unlock successive chapters, makes choices, adopts their persona, and completes light tasks tied to maps, instruments, and specimens.
4 • Connection	Across the story	Through historical characters and theory-of-mind moments, the visitor forms an emotional connection with Marsili and the people in his world — the Sense of Care payoff.
5 • Conclusion	Exit Hall	Marsili reveals that he donated his collection to found the Institute of Sciences and invites the visitor to “continue the mission.” The visit closes with reflection and a sense of completion.

This flow satisfies the institutional goal of attracting new young visitors (Moment 1 reaches them before they arrive), the authenticity goals (Moments 2–3 build mood and immersion), and the sense-of-care goals (Moment 4 builds emotional connection).

9. Summary: Decisions to Design

This brief is the resolved output of our CoDesignTool session. The table below maps each design stage to the decision it produced, so the whole team can see how play became product.

Stage	Resolved decision
Context	Audience = newcomers aged 14–27 with no prior knowledge of Marsili. Star assets = maps, Armenian dome, the historical city.
Institutional Goals	Higher visitor satisfaction (primary), more educational activities, more digitised assets.
Audience Goals	Authenticity — let time flow, create mood, build familiarity, strengthen personal context, stimulate the senses.
Design Brief	Motivations: time travel, entertainment. Barriers: poor signage, tech access. Capabilities: gaming, websites, web creation. Devices: smartphone, tablet, computer.
Ideation	Tech: audio, augmented authenticity, online. Engagement: decision, persona. Sense of Care: historical characters, enchantment, theory of mind.
Disruption	Mitigations for accessibility, attendance/connectivity, and cognitive balance.
Storyboard	Five moments: Discovery → Arrival → Exploration → Connection → Conclusion.